

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

**Week 2: Autoencoders: The Representation
Learning Model**

Today's Session Overview

- Regularization Techniques in AE: Contractive Autoencoders

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Contractive Autoencoders

A normal autoencoder says:

From the hidden representation, reconstruct the input well.

A **Contractive Autoencoder (CAE)** says:

Reconstruct the input well, but keep the hidden representation stable.

Consider:

original image: x and
slightly changed image: $x + \delta$

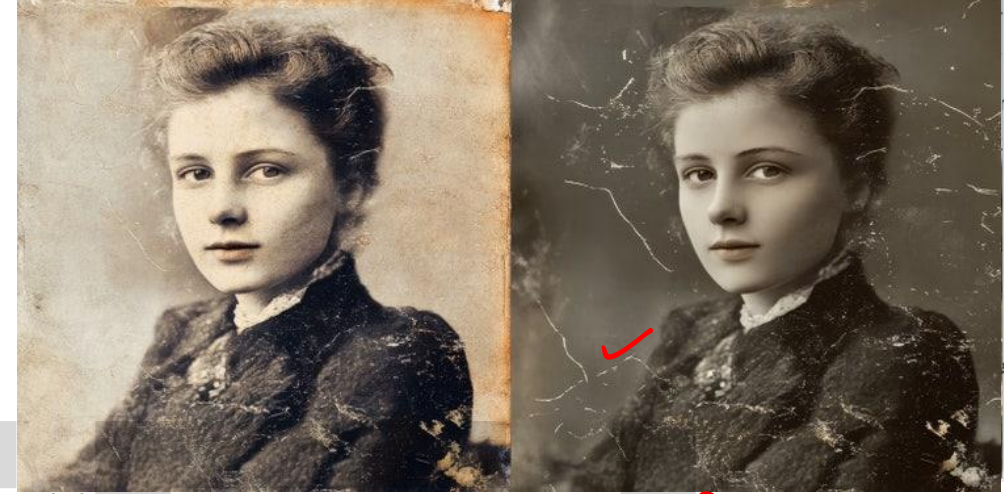
Where, δ means a **very small change**, such as a little noise, small brightness change, or one pixel variation.

A normal autoencoder may produce noticeably different hidden vectors for these two nearby inputs.

CAE tries to make

$$h(x) \approx h(x + \delta)$$

This means the hidden representation of the original image and the hidden representation of slightly changed image should remain nearly the same.

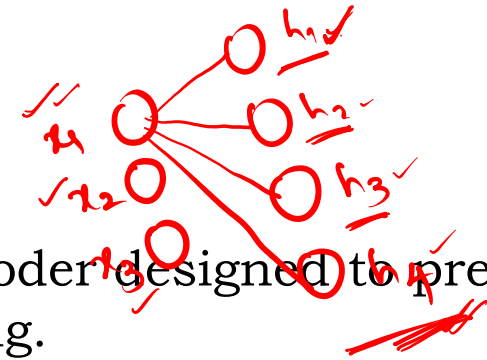


$x \rightarrow AE \rightarrow \hat{x}$

$x + \delta \rightarrow AC \rightarrow \hat{x}$

Stable means: When two input samples differ slightly, their hidden representations should also remain similar.

CAE goal =
good reconstruction + ✓
stable hidden representation

$$\begin{array}{c|c|c}
 x_1 & x_2 & x_3 \\
 \hline
 1.5 & 2.5 & 3.5 \\
 \hline
 1.6 & 2.5 & 3.4
 \end{array}$$


A contractive autoencoder is a regularized autoencoder designed to prevent an overcomplete autoencoder from learning a trivial identity mapping.

It is achieved by adding a Jacobian-based regularization term to the reconstruction loss.

$$\Omega(\theta) = \|J_x(h)\|_F^2$$

Here, $J_x(h)$ denotes the Jacobian of the hidden representation h with respect to the input x .

Here,

$$h = g(x)$$

A **CAE** tries to make these Jacobian values **small**, so the hidden representation becomes more stable.

- h is the hidden representation
- x is the input
- $J_x(h)$ contains all partial derivatives of the hidden units with respect to all input dimensions.

If the input has n dimensions and the hidden layer has k dimensions then

The first column contains the partial derivatives of all hidden units with respect to the first input feature x_1

$$J_x(h) = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_k}{\partial x_1} & \dots & \frac{\partial h_k}{\partial x_n} \end{bmatrix}$$

$$\|J_x(h)\|_F^2 = \sum_{j=1}^n \sum_{l=1}^k \left(\frac{\partial h_l}{\partial x_j} \right)^2$$

The squared Frobenius norm is the sum of the squares of all elements of the Jacobian matrix.

A **large Jacobian value** means the hidden representation changes a lot even for small input changes.

It measures how sensitively each hidden unit changes for a small change in the input.

$\Omega(\theta) = \|J_x(h)\|_F^2$ Jacobian-based regularization term

$$\mathcal{L}_{CAE} = \mathcal{L}_{rec}(x, \hat{x}) + \lambda \|J_x(h)\|_F^2$$



$\mathcal{L}_{rec}$ - Keep enough information so the model can reconstruct the input well.

$\Omega(\theta)$ - Do not let the hidden representation change too much.

∴ **Trade-off is to preserve only the important variations in the data while suppressing unimportant variations.**

The model cannot keep every detail and be completely insensitive at the same time. So it learns a balanced hidden representation.

The balance is controlled using the regularization parameter λ : How strongly should we punish the hidden representation for changing too much?

- Higher λ : more stable features.
- Lower λ : more detailed reconstruction.

If the input changes in a genuine and meaningful way that reflects the true pattern of the dataset, the model should notice that change. Otherwise, it cannot reconstruct the input properly. If the model ignores such a change, the reconstruction loss increases.

If the input changes mainly because of noise, random disturbance, or tiny perturbations, these changes are usually not important for reconstruction. If the model still reacts to them, the hidden representation changes even for small input changes, which increases the derivative values and hence the Jacobian penalty increases. Since these variations are not meaningful compared to the original image, the contractive penalty encourages the model not to respond strongly to them.

Next Session

- Numerical example
- Limitations of AEs

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Thank you